

Getting Started with VAFT

VAFT Introductory Tutorial | Session 01

Why this session exists

- ▶ Read a tokamak shot as a time-resolved experiment.
- ▶ Find diagnostic data in an OMAS/IMAS-style IDS structure.
- ▶ Make and question a plot through VAFT's public API.

A shot is a time-resolved experiment

1. Prefill and coil preparation establish initial conditions.
2. Breakdown and current rise mark the start of plasma formation.
3. A sustained-current interval supports diagnostic interpretation.
4. Ramp-down and shutdown complete the discharge.

A trace suggests questions about these phases; it is not a complete physical explanation on its own.

Diagnostics give complementary views

- ▶ Magnetic measurements track plasma current, fields, and flux.
- ▶ PF-coil currents document the programmed actuator response.
- ▶ UV spectroscopy adds a time-resolved impurity-emission signal.

Compare timing first. Use channel meaning, calibration, and other diagnostics before making a causal claim.

From IDS to a plot

VEST shot → OMAS ODS → IDS root → documented path →
VAFT public plot recipe

Examples: `magnetics`, `pf_active`, `spectrometer_uv`, and `equilibrium`.

The reproducible offline route

1. Load VAFT's packaged sample ODS for shot 39915.
2. Inspect IDS roots and the `magnetics.time` and `magnetics.ip.0.data` paths.
3. Discover recipes with `vaft.omas.available_plots`.
4. Plot plasma current, a magnetic overview, and UV intensity.

The guided lab route

Set `VAFT_TUTORIAL_MODE=lab` after configuring HSDS with `hsconfigure`. The notebook then performs a read-only magnetic-data selection with `vaft.database.open`.

No credentials or network are required for the offline lesson. An unavailable lab connection is a clear skipped extension, not an error in the core work.

Ask a defensible first question

- ▶ Where does plasma current rise, persist, and return toward zero?
- ▶ Which magnetic or spectrometer feature occurs near that transition?
- ▶ What metadata or additional diagnostic would strengthen the interpretation?

Independent exercise

Choose a supported plot recipe other than the guided plasma-current trace. Resolve its exported `vaft.omas.plot_*` function, then record:

1. the diagnostic and time interval;
2. one observed feature; and
3. one missing piece of evidence needed for interpretation.

Next: operation and vacuum fields

Session 02 connects these first diagnostic traces to the programmed discharge scenario, PF-coil evolution, and vacuum-field interpretation.